

What's driving growth of analysers?

Common factors	➤ Active electronically steered RADAR with multiple capabilities using complex signals ➤ New wireless standards 3G to 4G now moving towards 5G ➤ Smart antenna technologies like MIMO, Beam forming ➤ Complex smart phones with multiple technologies working simultaneously ➤ Study of electrical and magnetic properties of material
Power analyser	➤ Measurement of product efficiency for inverters, LEDs and electronic ballasts, UPS systems and transformers
Logic analyser	➤ Design, analysis and troubleshooting of complex digital or logic circuits ➤ Analysis of embedded designs with features such as advance trigger functions and serial bus decoding features like PC, SPI, CAN, UART, RS232
Spectrum analyser	➤ Receivers/transmitters -frequency response, bandwidth, modulation, harmonics ➤ Amplifier characteristics, filter characteristics ➤ EMI pre-compliance measurements ➤ Analysing RF spectrum ➤ Testing of active and passive equipment ➤ Finding of VSWR of DUT ➤ Spectrum monitoring and management ➤ Interference hunting ➤ EMC pre-compliance testing ➤ Capturing intermittent events

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real-time capabilities and get into trouble after getting the instrument.

Talking particularly about power analysers, Manisha Motiwale, technical manager, Scientific Mes-Technik Pvt Ltd, informs, "Manufacturers normally specify basic accuracy, but user should consider the accuracy specification for the complete range in which measurement is done. Also, one should check the standard and optional features, and accessories as per requirement."

Why real-time processing

In most cases, what usually happens is that the test engineers are told, "You have to test this particular protocol coming from your system. Get an analyser for that," informs Mohanram. He says, "That's the mandate an engineer gets. Sure there are a lot of analysers which can analyse that particular protocol, but what they don't share is the information on how they do that

analysis." These analysers capture all the information and a software on the instrument takes that data and processes it off-line to give the analysed information.

In the above case, if a particular protocol needs handshaking, it becomes completely impossible for people to use those off-line analysers. Handshaking means, based on an incoming signal, the analyser has to generate some other signal to establish communication and, once that happens, it starts pumping out data. In such cases, it is not feasible to use off-line analysers. To avoid this, engineers should look for online analysers.

Off-line analysis ideally has a transmitter and receiver, which can only split packets and see what is happening, Mohanram notes. "That is not a complete test that you are doing, because if you are trying to analyse a packet or trying to analyse commu-

nication protocol, you would want to do it as a different layer like physical or application layer, for instance. But if you just do splitting then you are not able to completely test the capabilities at every different layer. That drawback can be overcome by taking an analyser that can do in-line processing."

Pricing

For almost the same cost that people used to pay for box type analysers, the high-end FPGA-enabled analysers come in really handy. "I would say, at the same cost there is more and more functionality getting packed into these analysers and they are shrinking in size," says Mohanram.

Using a PC to do the processing work for all manner of test and measurement equipment may not be feasible for the larger manufacturers, but it does serve the user very well. "PC-based processing for test equipment is here to stay," says Bruce Devine, CEO, Test Equipment Plus, Inc. It puts pricing pressure on the traditional companies that build test equipment with the processor inside their equipment. He adds, "It is now difficult for many companies to justify the expense of buying anything but PC-based test equipment because of the cost savings, upgradability and extensibility of PC-based systems."

Today's analysers are a boon for engineers

Design engineers are supposed to spend time on designing their products rather than learning new instruments and spending time debugging instrument related issues. Today's instruments cater to all the different issues generally faced by these engineers. They are reliable, can be easily automated and can be reused for different platforms. Various software tools support instruments covering different technologies. Thanks to these capabilities, they enable engineers to invest least possible time on measurement, be confident about the measurements and spend more time on designing efficient products. ■

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